

Design Space Exploration of Backside Clock Meshes for 2 nm GAAFET BSPDN Technology

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Abstract—Clock meshes are used in high-performance VLSI designs to minimize skew and tolerate on-chip variation, but they spend scarce routing resources on premium metal layers. Backside power delivery creates a new option: it adds thick, low-resistance metal layers on the back of the wafer, and the power grid does not consume all of them. Flip-flops remain on the frontside; a backside mesh therefore cannot drive them directly, and every connection passes through a through-silicon via. We present the first design-space exploration of backside clock meshes, implemented in OpenROAD on GT2N, a 2 nm nanosheet technology. Four benchmarks (1,938 to 15,311 flip-flops) are explored with multi-objective Bayesian optimization, and every design point is verified by transistor-level SPICE simulation, since the cyclic mesh cannot be evaluated by static timing analysis. Across all four designs, the backside mesh consistently outperforms an identical frontside mesh, with on average 45% lower skew, 25% lower sink slew, 4.5% lower power, and 28% less frontside clock wiring, and its skew spread under 10,000-sample Monte Carlo is a third of the frontside mesh’s.

I. INTRODUCTION

A clock mesh consists of a grid of wires distributed across the die and driven from multiple points. Since thousands of sink paths share the same grid, local delay differences are averaged out instead of accumulating along individual paths. This results in very low skew and good tolerance to process and environmental variations, which is why clock meshes are used in high-performance designs [1]–[4]. There also has been research efforts regarding their fast analysis for optimization [5], [6]. One of the main drawbacks in clock meshes are power and routing overhead: the entire grid capacitance must be charged every cycle, while the grid also occupies routing tracks on the metal layers that are needed for signal routing and the power grid.

The power grid is itself a heavy consumer of those metal layers, and backside power delivery (BSPDN) removes it from the frontside entirely. Buried power rails beneath the standard cells are fed from a power grid on the back of the wafer, eliminating the frontside power straps and improving IR drop at the same time [7]–[9]. The approach is now in production at Intel and is on the TSMC and Samsung roadmaps [10]–[12]. More importantly for a clock mesh, BSPDN adds backside metal layers that are wider and thicker than the frontside layers a mesh would normally occupy, and the power grid does not use all of their tracks. In the 2 nm GT2N technology we target, the backside metals have $3.5\times$ lower resistance per

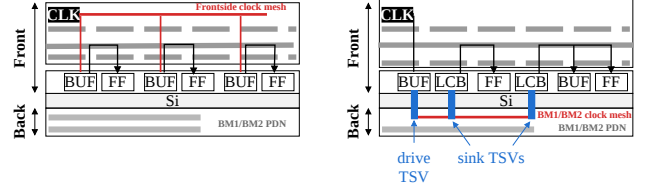


Fig. 1: Moving the mesh to the backside forces every mesh-to-sink connection through a TSV and cuts a buried power rail at every landing.

micron than the M6–M9 class and $22\times$ lower than M4/M5 (Section II-A). This low wire resistance is what makes a backside clock mesh worth investigating.

Prior work has begun to exploit backside metal for clocking, but majorly for clock *trees*. Veloso et al. route the global H-tree portion of a high-performance CPU’s clock on the backside metals and report a clear latency reduction from even that partial move [9]; Bethur et al. select which tree nets to flip by fanout or by a GNN-predicted timing criticality [13], [14]; Vanna-iampikul et al. share the backside metals between the power grid and selected clock trunk nets [15]; Jiang et al. synthesize the double-sided tree directly with concurrent buffer and nTSV insertion [16]; and CSCO allocates the limited backside tracks jointly between clock and signal nets [17]. To our knowledge, backside clock meshes have not yet been explored.

A mesh cannot simply be relocated (Fig. 1). The backside layers already carry the power grid, so the mesh must fit around it. The flip-flops remain on the frontside, so every mesh-to-sink connection must cross the wafer through a through-silicon via (TSV). Each TSV passes through a buried power rail, which must be cut around it; and the shorted grid is a cyclic network that static timing analysis cannot evaluate, so every candidate configuration must be simulated at the transistor level. The question is whether the mesh still wins once the cost of reaching it from the frontside is paid.

This paper makes the following contributions:

- Upstream OpenROAD support for backside routing, buried-power-rail cutting at TSV landings, and backside IR analysis, which the flow of this paper builds on.
- The first construction flow for backside clock meshes,

implemented in OpenROAD on the open-source GT2N 2 nm nanosheet BSPDN technology.

- A multi-objective Bayesian design-space exploration over four mesh parameters, with every design point verified by full-network HSPICE simulation.
- A comparison against two frontside mesh baselines on four benchmarks (1,938 to 15,311 flip-flops), with 10,000-sample Monte Carlo analysis.

Across all four designs, the evaluated backside mesh configurations achieve lower clock skew, lower maximum sink slew, and lower frontside clock wirelength than the corresponding frontside mesh configurations, and are markedly less sensitive to variation. Section II describes the technology and the construction flow, Section III the exploration method, Section IV the setup, and Section V the results.

II. BACKSIDE CLOCK MESH SYNTHESIS

The challenges outlined above affect every step of constructing a backside clock mesh. This section first describes the target technology and then presents the construction flow: placing the mesh on the backside tracks with constrained resources after power grid construction, crossing between the backside and frontside through TSVs and buried power rails, and connecting the mesh to the frontside sinks through local clock buffers.

A. Target Technology

We build on GT2N, an open-source 2 nm nanosheet technology with buried power rails and backside power delivery [18]. GT2N provides fourteen frontside metal layers (M0–M13), tungsten buried power rails beneath the standard cells, and four backside metals: BM1/BM2 at 56 nm width and ultra-thick BM3/BM4 at 360 nm [18]. The low resistance of the backside metals makes them well suited to clock meshes: based on the PDK’s extracted wire resistances, BM1/BM2 are $3.5\times$ less resistive per micron than the M6–M9 class (7.5 vs. $26\ \Omega/\mu\text{m}$) and $22\times$ less resistive than M4/M5. Crossings between the frontside and backside are implemented using TSV cells placed in the standard-cell rows and modeled as a $6\ \Omega$ series resistance [13]. The cell library provides two nanosheet widths (13 and 31 nm) and five threshold-voltage options.

B. Mesh Construction over a Backside PDN

Power reaches the standard cells through buried power rails beneath each cell row, which are fed by a power grid on BM1 and BM2. We use these same layers for the clock mesh, placing horizontal wires on BM2 and vertical wires on BM1 in the tracks left available by the power grid. The power grid is built first, and mesh wires that overlap or are too close to power straps are shifted to the nearest clear track or removed if no valid track exists. Each remaining intersection receives a via, forming a single electrically shorted mesh net.

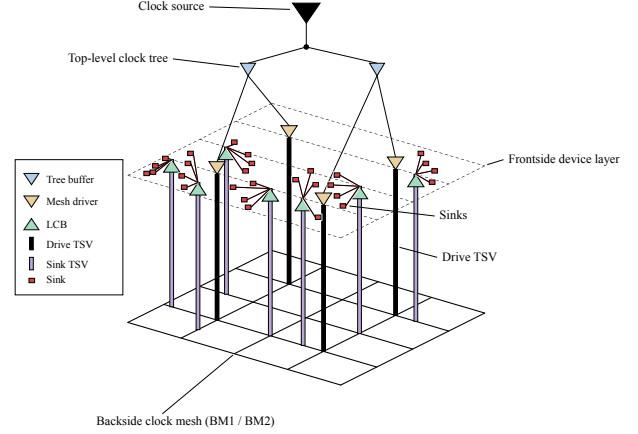


Fig. 2: The clock passes through three buffering stages (tree, mesh drivers, LCBs) and crosses between frontside and backside through drive TSVs and sink TSVs.

C. Clocking Architecture and Buffering

The mesh itself is passive; the clock reaches it and continues to the sinks through four active stages (Fig. 2). A conventional buffer tree synthesized by TritonCTS [19] distributes the clock from the root to the mesh driver inputs. Each driven grid intersection holds one mesh driver, a standard buffer placed on the frontside directly beneath its grid node and snapped to the nearest cell row, whose output crosses to the backside through a TSV (Section II-D). Drivers occupy every other intersection in a checkerboard, which halves the driver and TSV count without changing the wire grid. Below the mesh, sink connections return to the frontside through per-cluster TSVs and local clock buffers (LCBs, Section II-E). The tree needs no precise balancing; it drives only tens to a few hundred driver inputs, and the shorted mesh averages whatever arrival differences it delivers.

All drivers on a mesh use the same buffer cell, whose drive strength is an explored parameter (Table I). The exploration consistently prefers small drivers (x1–x4), even on the largest design. The LCB tier (Section II-E) is sized independently; its role is to regenerate the clock edge for a small group of flip-flops.

D. TSV Insertion and Buried Power Rail Breaking

Two kinds of TSV are used: a drive TSV at each mesh driver and a sink TSV at each LCB (on Ibex at a $5\ \mu\text{m}$ pitch, 41 and 382 respectively). A clock TSV uses the same via-stack structure that connects the buried power rails to the backside grid, but carries the clock net through the BPR layer. The rail there would short to it and violate the BPR spacing rule, so the flow cuts the rail at every landing, removing a piece sized by the TSV footprint plus a $0.224\ \mu\text{m}$ clearance on each side, and deletes any rail fragments left floating between nearby cuts; on the largest design this amounts to about 2,500 cuts. Power delivery is unaffected, since each cut is well under a

micron and the remaining rail pieces are still fed through their own via stacks. Cells displaced by the cuts are re-legalized. None of this is needed for the frontside meshes compared in Section V.

E. Local Clock Buffers and Sink Clustering

Because the flip-flops are frontside cells, giving each its own TSV would require thousands of crossings. Instead, flip-flops are grouped into clusters of at most F_{fanout} sinks: candidate taps are generated along the mesh wires, and each flip-flop joins the nearest tap whose cluster has not reached the fanout bound F_{fanout} . Each cluster receives one LCB, legalized with its neighbors, and one sink TSV to the LCB input; the LCB output drives its flip-flops as an ordinary routed net. On Ibex with $F_{\text{fanout}} = 6$ this yields 382 LCBs for 1,938 flip-flops. The tier also limits the mesh load to a few hundred buffer inputs. For the frontside baselines in Section V, the same tier connects each LCB to the nearest point on the frontside mesh without TSVs, and a second variant omits the tier and attaches flip-flops directly to the mesh wires.

III. DESIGN-SPACE EXPLORATION

TABLE I: Four parameters interact enough that a per-design joint search is required.

Parameter	Values
Mesh pitch (μm)	1.6–14 (per-design range)
Mesh driver size	x1, x2, x3, x4, x6, x8, x10, x12
LCB fanout bound F_{fanout}	6, 8, 16, 24, 40, 80
LCB size	x1, x2, x3, x4, x6, x8, x10, x12

The flow leaves four parameters open (Table I), which define a few thousand configurations per design, and each evaluation requires the full flow of Section II plus HSPICE simulation, taking 10–50 minutes per point. Exhaustive search is out of reach, and one-at-a-time sweeps are misleading because the parameters interact: whether a 10 μm pitch is usable depends on the LCB fanout bound, since small clusters keep the sparse mesh’s edges sharp while large ones push them past the slew limit.

We therefore search all four parameters jointly with the multi-objective Tree-structured Parzen Estimator (MOTPE) [20], [21] in Optuna [22], minimizing worst-case sink skew and clock network power, following the use of Bayesian tuning in physical design by AutoTuner, FIST, AutoDMP, and BOOM-Explorer [23]–[26]. TPE handles our categorical parameters (buffer cells, discrete pitches) natively, and its multivariate mode learns which combinations work rather than scoring each parameter independently [25].

Because skew and power trade against each other, we retain the full non-dominated set. A trial is reported as infeasible when detailed routing fails, when any flip-flop is missing or fails to toggle, or when the worst sink slew exceeds 10% of the clock period; infeasible outcomes still steer the sampler away from those regions.

IV. EXPERIMENTAL SETUP

All experiments run in OpenROAD with our clock-mesh extensions on the GT2N technology of Section II-A. GT2N provides libraries and device models at a single corner (TT, 0.7 V, 25 °C), so nominal results are reported at that corner; every skew, slew, and power number comes from transistor-level HSPICE simulation of the complete clock network.

We evaluate four benchmarks (Table III): the Ibex RISC-V core and a JPEG encoder from OpenROAD Flow Scripts [27], and the Minimax bit-serial RISC-V CPU and the FloopNoC network-on-chip from the HighTide suite [28]. Each is synthesized and placed with the standard OpenROAD flow before clock construction, so sink locations are identical for every clock network compared on a design. Backside power straps are as wide as the design rules allow and numerous enough to carry each design’s current, verified with OpenROAD’s IR analysis. The exploration of Section III runs 300–500 trials per design, three in parallel, with completed configurations cached; most infeasible trials fail the slew bound at sparse pitches.

A. Variation Model

Monte Carlo runs 10,000 HSPICE samples per network on Ibex under the parameters of Table IV. Device σ follow N3 nanosheet projections [29] and wire σ follow [30]; nominal values are GT2N’s, and every device and wire element is drawn independently.

V. RESULTS

A. Pareto Frontiers

Fig. 3 shows the skew–power frontier found on each design. All four share the same shape: skew falls steeply as power rises from the sparsest feasible pitch. The representative points in Table II are taken at the knee, and each frontside baseline is built with the identical grid, pitch, and driver configuration.

B. Backside vs. Frontside

Against the frontside mesh with the same LCB tier (FS+LCB), the backside mesh lowers worst-case skew on every design, by 20% on JPEG and by 49–57% on the other three, 45% on average, and sharpens the worst sink slew by 16–38%. It does so at 1.5–6.5% lower clock power, thanks to the backside grid’s lower resistance, and with 20–40% less frontside clock wirelength, because the grid itself has left the frontside. Latency rises by 2–10 ps on three designs, the cost of the extra TSV and LCB stage, and falls on Minimax. The direct-attach frontside mesh (FS Direct) draws the least power but pays for it in slew: its worst sink edge is 2.6–4.7 $\times$ slower than the backside mesh and exceeds the 10% slew budget on Minimax, so it is not a usable design point at these pitches.

C. Power Delivery

Sharing BM1/BM2 with the clock mesh and cutting the buried power rails at the TSV sites keeps the supply within budget. On Ibex, worst-case VDD drop is 30.7 mV for the backside configuration against 32.2 mV for the frontside mesh,

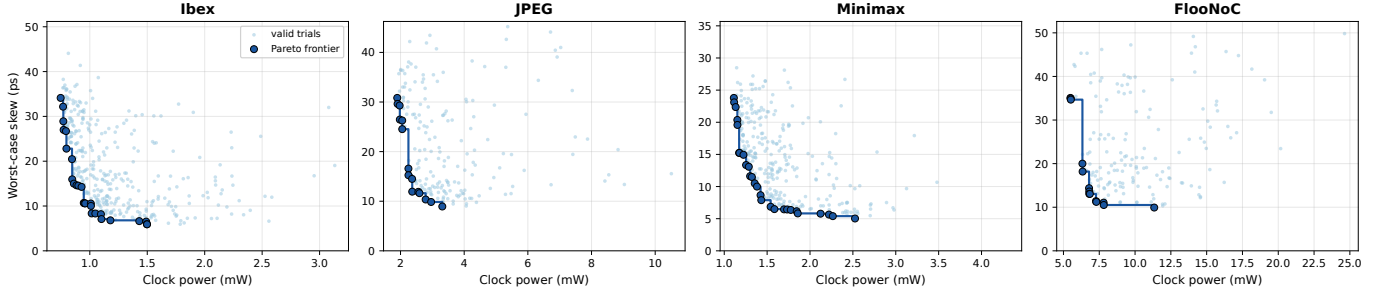


Fig. 3: Skew–power Pareto frontiers of the backside clock mesh. Light points are all feasible trials; dark points form the non-dominated set.

TABLE II: The backside mesh wins on skew, slew, power, and frontside wire on every design at a small latency cost. BS+LCB = backside mesh + LCB tier; FS+LCB = frontside mesh + LCB tier; FS Direct = frontside mesh with direct flip-flop attachment. Checkerboard mesh buffers; full-network SPICE.

Benchmark	Ibex			JPEG			Minimax			FlooNoC		
	BS+LCB	FS+LCB	FS Direct	BS+LCB	FS+LCB	FS Direct	BS+LCB	FS+LCB	FS Direct	BS+LCB	FS+LCB	FS Direct
Mesh grid	9×9	9×9	9×9	25×25	25×25	25×25	9×9	9×9	9×9	20×20	20×20	20×20
Pitch (μm)	5	5	5	3.2	3.2	3.2	4	4	4	6	6	6
Mesh buffers	41	41	41	313	313	313	41	41	41	200	200	200
Local clock buffers	382	382	N/A	790	790	N/A	135	135	N/A	2,611	2,611	N/A
#TSVs	423	N/A	N/A	1,103	N/A	N/A	176	N/A	N/A	2,811	N/A	N/A
Clock skew (ps)	6.80	15.28	29.78	9.85	12.34	19.61	20.36	40.00	39.4	11.05	25.76	52.27
Max sink slew (ps)	14.1	19.1	53.2	19.9	23.8	57.8	47.5	76.4	125.6 [†]	23.6	29.7	111.3
Clock power (mW)	1.179	1.261	0.806	2.961	3.006	2.088	1.153	1.220	1.053	7.804	8.184	5.256
FS clock WL (μm)	2,159	3,055	3,031	6,008	10,025	9,017	1,896	2,476	2,753	17,678	22,214	23,995
Clock latency (ps)	51.5	49.3	57.0	69.5	60.1	70.4	56.8	58.5	88.9	92.0	88.7	112.3

[†]violates the slew budget (10% of the clock period).

TABLE III: Benchmark designs: two RISC-V cores, a JPEG encoder, and a network-on-chip, from 1,938 to 15,311 flip-flops.

Design	Insts	FFs	Core (μm)	T_{clk} (ns)
Minimax	25,071	2,251	35.6 × 35.6	0.78
Ibex	24,117	1,938	44.0 × 43.9	1.00
JPEG	104,562	4,384	80.6 × 80.5	1.00
FlooNoC	122,079	15,311	117.0 × 116.9	1.15

TABLE IV: Monte Carlo variation parameters (Gaussian, 1σ listed). Device σ follow N3 nanosheet projections [29] and wire σ follow [30]; nominal values are GT2N's [18].

Parameter	μ	σ	From
V_{th}	177 mV	20 / 15 mV*	[29]
L_g (gate length)	14 nm	0.167 nm	[29]
W_{NS} (sheet width)	31 nm	0.167 nm	[29]
T_{NS} (sheet thickness)	5 nm	0.133 nm	[29]
V_{dd}	0.7 V	18 mV [§]	IR drop
Wire R (FS)	extracted	12.5% [¶]	[30], [31]
Wire R (BS)	extracted	6.25% [¶]	scaled
Wire C	extracted	3%	[30]

*pMOS/nMOS. [§]Worst-case backside IR drop taken as a 3σ bound. [¶]Frontside σ is above the 5.6–9.9% measured at 11 nm [30] because narrower 2 nm wires are more sensitive to surface and grain-boundary scattering [31]; no backside data is published, so the wider 56 nm backside lines are assigned half the frontside value.

and VSS drop rises from 16.8 to 22.5 mV; both are within the budget used to size the straps in Section IV.

D. Skew Under Variation

TABLE V: Monte Carlo clock skew on Ibex (10,000 HSPICE samples per network, parameters from Table IV).

	BS+LCB	FS+LCB	FS Direct	CTS tree
Nominal skew (ps)	6.77	15.20	29.82	32.43
μ_{skew} (ps)	7.13	15.44	30.02	34.40
σ_{skew} (ps)	0.34	0.99	0.78	1.79
$\mu + 3\sigma$ (ps)	8.14	18.43	32.35	39.77
Max skew (ps)	8.57	19.24	32.82	42.44

Table V reports Monte Carlo results on Ibex, with a conventional TritonCTS [19] tree added as a fourth network. The backside mesh has the smallest mean skew and the smallest spread: its σ_{skew} of 0.34 ps is a third of the frontside mesh's and a fifth of the tree's.

VI. CONCLUSION

We have presented a construction flow and design-space exploration for clock meshes on the backside metals of a BSPDN technology. Building the mesh around the backside power grid, reaching it through TSVs that cut the buried power rails, and returning to the frontside sinks through a local clock buffer tier are all workable in OpenROAD on GT2N. With every design point verified in SPICE, the backside mesh beats an identical frontside mesh on all four benchmarks in skew, slew, power, and frontside wiring, and holds that lead under process and supply variation.

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